CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First semester of B. Tech. Examination (CE/ME/EE/CL/IT/EC)

May 2013

MA-101, Engineering Mathematics-I

Date: 10/05/2013 Time: 1.30 p.m to 4.30 p.m Maximum Marks: 70

INSTRUCTIONS:

(a) The question paper comprises of two sections.
(b) Section I and II must be attempted in separate answer sheets.
(c) Marks suitable assumptions and draw neat figures wherever required.
(d) Use of scientific calculator is allowed.

Section-I

Q. 1(a) Whether the following statements are true or false? [05]

(1) If determinant of a matrix is zero, then inverse of that matrix is not exist.
(2) In the row-echelon form of the matrix, the number of non-zero columns is the rank of a matrix.
(3) The Leibnitz's test can be apply only for alternate series.
(4) The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is convergent.
(5) The modulus of complex number $3 + 4i$ is 5.

(b) Give the necessary and sufficient condition for system $AX = B$ of non-homogenous linear equations to be consistence. [02]

Q.2(a) Define rank of a matrix and find rank of the matrix $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$ using minors. [06]

(b) Find inverse of a matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ by Gauss-Jordan method, if exist. [05]

(c) Discuss the convergence of the series $\sum_{n=0}^{\infty} \frac{1}{2^n}$ and if it is convergent find its sum. [03]

OR

Q.2(a) Solve the equations [05]

5x+3y+4z=0,
2x-y+z=0,
3x+y+2z=0.

(b) Convert the matrix $\begin{pmatrix} 0 & 2 & 1 \\ 5 & 2 & 3 \\ 1 & 1 & 0 \end{pmatrix}$ in row-echelon form and also find its rank and nulity. [05]

(c) Discuss the convergence of the series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n\sqrt{n}}$. [04]

Q.3(a) State De Movire's theorem and hence evaluate $(1+i\sqrt{3})^{90} + (1-i\sqrt{3})^{90}$. [06]

(b) Solve: $z^4 + 1 = 0$. [04]

(c) Discuss the convergence of the series $\sum_{k=0}^{\infty} \frac{1}{k(k+1)}$ and if it is convergent find its sum. [04]

OR

Q.3(a) If α and β are roots of equation $z^2 \sin^2\theta - z \sin 2\theta + 1 = 0$, prove that $\alpha^n + \beta^n = 2 \operatorname{cosec}^n \theta \cos n\theta$, where n is integer. [06]

(b) If w is complex root of unity then prove that $1 + w + w^2 = 0$. [04]

(c) Show that the series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges for all x. [04]

Section-II

Q.4(a) Find the n^{th} derivative of $\cos^2 x$. [01]

(b) Using Maclaurin's series expand $ln(1+x)$ upto four terms.. [03]

(c) If $u = sin^{-1}\left(\frac{x}{y}\right) + tan^{-1}\left(\frac{y}{x}\right)$, then find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$. [03]

Q.5(a) If $u = \sin^{-1}\frac{x^2+y^2}{x+y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$. [04]

(b) Evaluate: $\lim_{x\to\frac{\pi}{2}} \frac{\tan x}{\tan 3x}$ [02]

(c) The deflection at the center of a rod of length l and diameter d supported at its ends and loaded at the center with a weight w varies as wl^3d^{-4}. What is the percentage increase in the deflection corresponding to the percentage increase in w, l and d of 3,2 and 1 respectively? [04]

(d) If $x = r\cos\theta$, $y = r\sin\theta$, evaluate $\frac{\partial(r,\theta)}{\partial(x,y)}$ and $\frac{\partial(x,y)}{\partial(r,\theta)}$. [04]

OR

Q.5(a) Given $u = xy + xz + yz, x = r, y = r\cos t, z = r\sin t$, find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial t}$. [04]

(b) Evaluate: $\lim_{x\to 0} x^x$ [02]

(c) Maximise $x^2y^3z^4$ subject to $2x + 3y + 4z = 9$. [04]

(d) If $z = \sec^{-1}\frac{x^3+y^3}{x+y}$, show that $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2\cot z$. [04]

Q.6(a) Find approximation: $\sin 29^\circ \cos 58^\circ$. [04]

(b) If $u = 2x^2 - yz + xz^2$, where $x = 2\sin t, y = t^2 - t + 1, z = 3e^{-t}$, find $\frac{du}{dt}$ at $t = 0$. [02]

(c) Find the equation of the tangent plane and normal line to the surface $2x^2 + y^2 + 2z = 3$ at $(2, 1, -3)$ [04]

(d) Test the relative maxima and minima for the function $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$. [04]

OR

Q.6(a) Find the equations of tangent plane and normal line to the surface $x^3 + 2xy^3 - 7x^2 + 3y + 1 = 0$ at $(1, 1)$. [04]

(b) Find total differential of $w = xe^{2y} + e^{-y}$. [02]

(c) If $u = x^2$, $v = y^2$, evaluate $\dfrac{\partial(u, v)}{\partial(x, y)}$. [04]

(d) Find approximation: $\sqrt{(299)^2 + (399)^2}$. [04]

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